THE UNIVERSITY OF SYDNEY

State Aware Risk Controller (STARC) for the Stopping Decision in Agentic Systems

Harry Kember

Supervisor: Prof. Ying Zhou

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AS MORE SYSTEMS BECOME AGENTIC, A NATURAL PROBLEM IS WHEN TO STOP REASONING

AgenticLU refines its answers over multiple rounds

Instead of generating an answer in one pass, the system clarifies, gathers evidence and revises across multiple rounds.

More rounds are not free, and not always better

Each round costs computation, and can entrench a wrong reasoning path or even push a correct answer to wrong

Despite this, the system never decides if more is required

It clarifies to fixed depth. Other systems stop on model confidence. Neither is adequate.

So when should an agentic system stop reasoning?

AGENTICLU CLARIFIES OVER MULTIPLE ROUNDS, ASSUMING EACH ROUND HELPS

AgenticLU is the prior system this thesis builds on.

To handle unreliable use of evidence over long context, it answers a question through an explicit clarification round rather than in a single pass.

AgenticLU System Diagram for a Question

One full **round** of clarification, made up of 4 operations

1. **Generate clarification question**
   Forms a focused sub question from the original query
2. **Select relevant context**
   Pins context the system deems most relevant
3. **Answer clarification question**
   Produces an answer to clarification question from the pinned context
4. **Generate final answer**
   Answers the original question using the relevant context and clarification answer

ASSUMPTIONS MADE BY THE SYSTEM

- Every clarification round helps so it runs a fixed number of rounds, regardless of state.
- Model confidence signals when to stop: the same signal known to be unreliable.

EXISTING SIGNALS ARE UNRELIABLE IN AGENTIC SYSTEMS

Fluency misleads

A confident, well-formed answer can still be weakly supported by the evidence.

How good an answer looks isn't evidence it's right

Fluency can't signal when to stop

Confidence can't be trusted

Model confidence is miscalibrated in long-context QA. It doesn't track correctness, grounding, or any other signals.

Model confidence can't signal when to stop

None of these surface signals reveals when to stop

STARC ANSWERS THE STOPPING QUESTION FROM THE AGENTIC SYSTEM STATE NOT MODEL CONFIDENCE

Existing solutions

How confident is the model?

A signal known to be unreliable

STARC

State Aware Risk Controller

Does the agentic system state warrant further reasoning?

By reading the artifacts that produced the answer

The central claim: STARC composes LLM judgements with deterministic logic, producing a reliable stop / continue decision that performs across heterogenous question types

THE AGENTIC SYSTEM STATE MUST BE MADE OBSERVABLE TO STUDY CLARIFICATION

The AgenticLU runtime exposed only a final answer. A non-intrusive instrumentation layer records every artefact produced at each round boundary, leaving the clarification mechanism itself unmodified.

WHY IS THIS IMPORTANT?

- An internal computation becomes a structured object that can be **read, classified, and compared**.
- **It enables offline replay:** every policy is tested on the identical saved state, so any difference reflects the stopping logic: not variation in the underlying model.

THE EFFECT OF FURTHER CLARIFICATION DEPENDS ON THE FIRST ROUND STATE

Two case study examples

REPAIR: ambiguity resolved

Question:	*"Who is the guy that jumped from space"*
Round 1:	Names both Baumgartner and Kittinger commits to neither.
Round 2:	Asks "before or after 1960?" and commits to the correct answer.

The candidate was already in the first round state; clarification found it.

DEGRADATION: overspecified

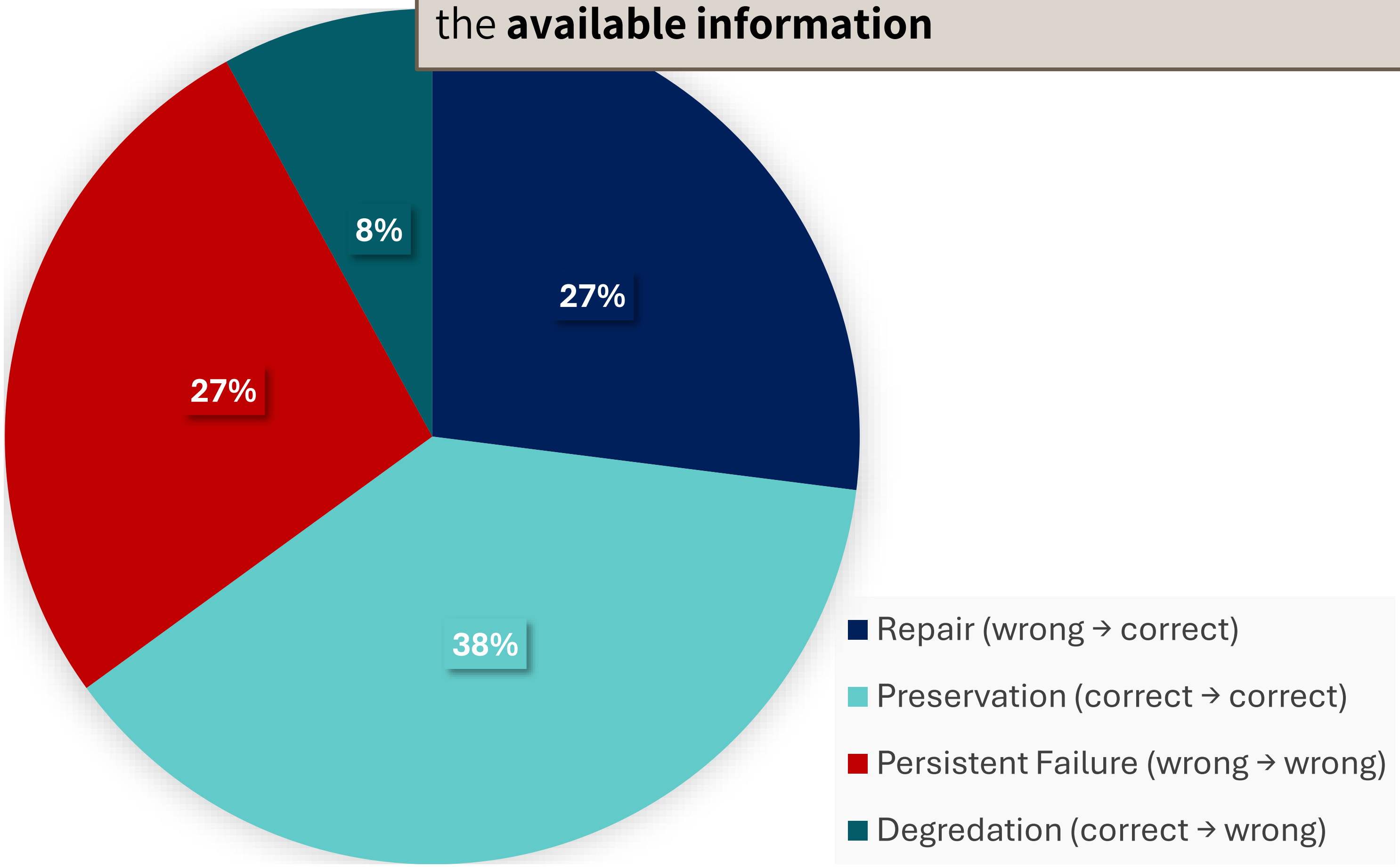
Question:	*"When did they stop making pull tabs on beer cans"*
Round 1:	Provides the correct answer: "the 1970s."
Round 2:	Injects an unsupported premise (1982 Act) and adopts it wholesale, giving the wrong answer.

A noncommittal first round answer met by a premise heavy question degraded.

Properties that indicate a well supported answer can be **determined by examining the system state at the boundary**, before the next round runs

FURTHER CLARIFICATION ONLY HELPS SOMETIMES. STOPPING IS A REAL DECISION

These results **motivate** the need for a controller that decides *stop* or *continue* based on the **available information**



PRESERVATION DOMINATES

In most instances, the second round preserves the answer the first round produced.

THE ASYMMETRY

Continuation rarely harms a correct answer but only repairs some incorrect ones.

NO TRIVIAL SOLUTION

A trivial solution *always stop* or *always continue* can be applied here.

THE STOPPING DECISION HAS FOUR OUTCOMES: THEY HAVE UNEQUAL COST

Correct Decision

Safe stop
Stop on a correct answer

Justified continue
Continue on a wrong answer

Wrong, but not as costly

Unnecessary continue
Continue on a correct answer

Premature stop
Stop on a wrong answer

Costly, irreversible mistake

The correctness of the decision is separated from the outcome of further clarification
A correct decision to continue does not mean that the second round of clarification will produce a better answer. This is out of scope.

The two errors are not equal, so the **rule must be asymmetric**

Stopping policies demands strong evidence to stop

STARC ANSWERS THE STOPPING DECISION BY EVLATING THE AGENTIC STATE

Implemented as three polices, each evaluating the state in a different way

VRB
Verification and Refinement Baseline

Dual perspective LLM judges read the agentic state

Verifier DEFENDS: does the evidence support the answer?
Falsifier ATTACKS: is there a competing answer or a reason to doubt it?

Outputs: stop / continue

VRRS
Verification and Refinement with ***Risk Signals***

Adds a deterministic layer over the saved state and VRB's judgements

Hard signals: concrete failure patterns *(multi-candidate answer, form mismatch)* that force continuation
Soft signals: weaker concerns that flag a case as borderline

Outputs: stop / continue / **borderline**

VRRS-BR
Verification and Refinement with Risk Signals with ***Borderline Review***

Adds a semantic reviewer, invoked **only** on borderline cases.

Reads the trace for a concrete, **identifiable error** *(a contradiction, a wrong entity, or a better competing answer)*

A final filter admits its verdict only when a real error is found: stop / continue

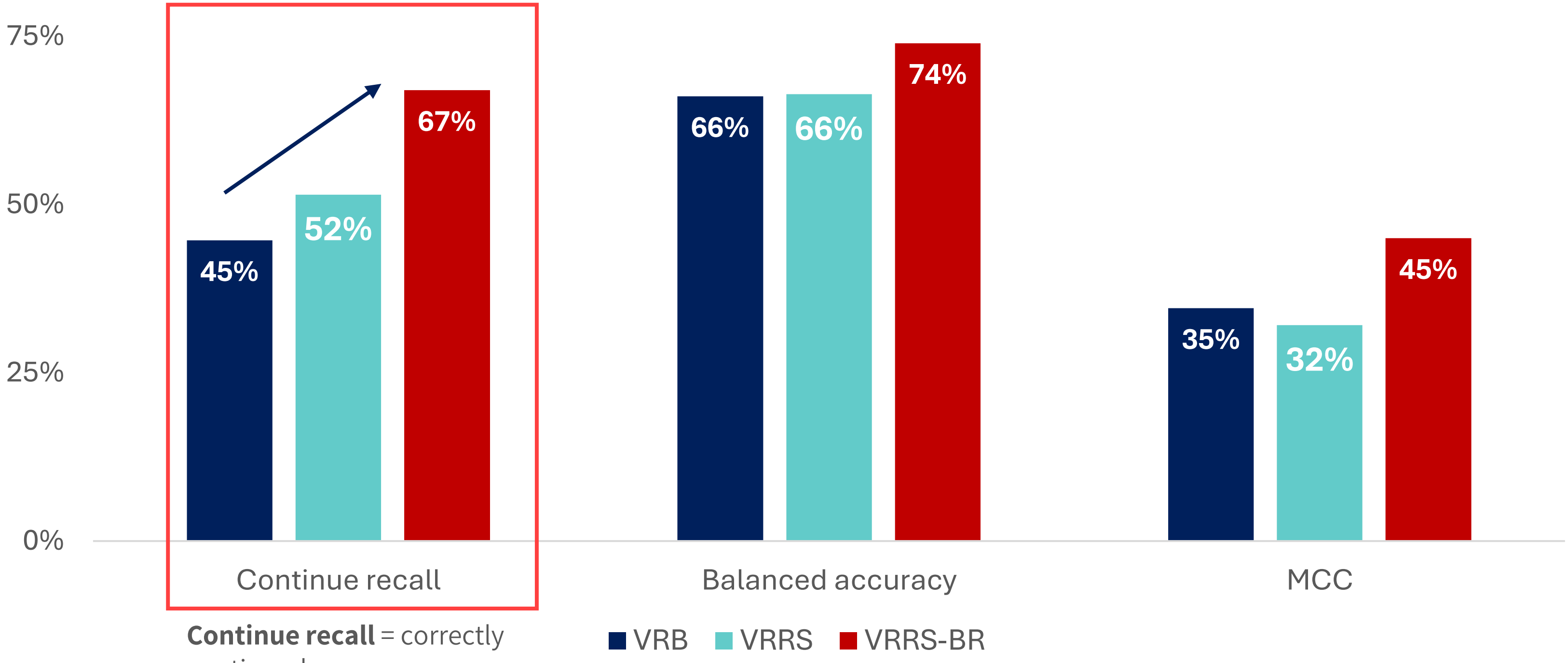
LLM judges the trace but **never decides alone**

Deterministic logic composes its judgements into the call

No single confident output is trusted on its own

THE BEST POLICY IS VRRS-BR, CATCHING 67% OF WRONG ANSWERS, UP FROM 45%

A risk sensitive redistribution, not a just uniform efficiency gain.



Continue recall = correctly continued on wrong answers

CONTROLLERS TRADE

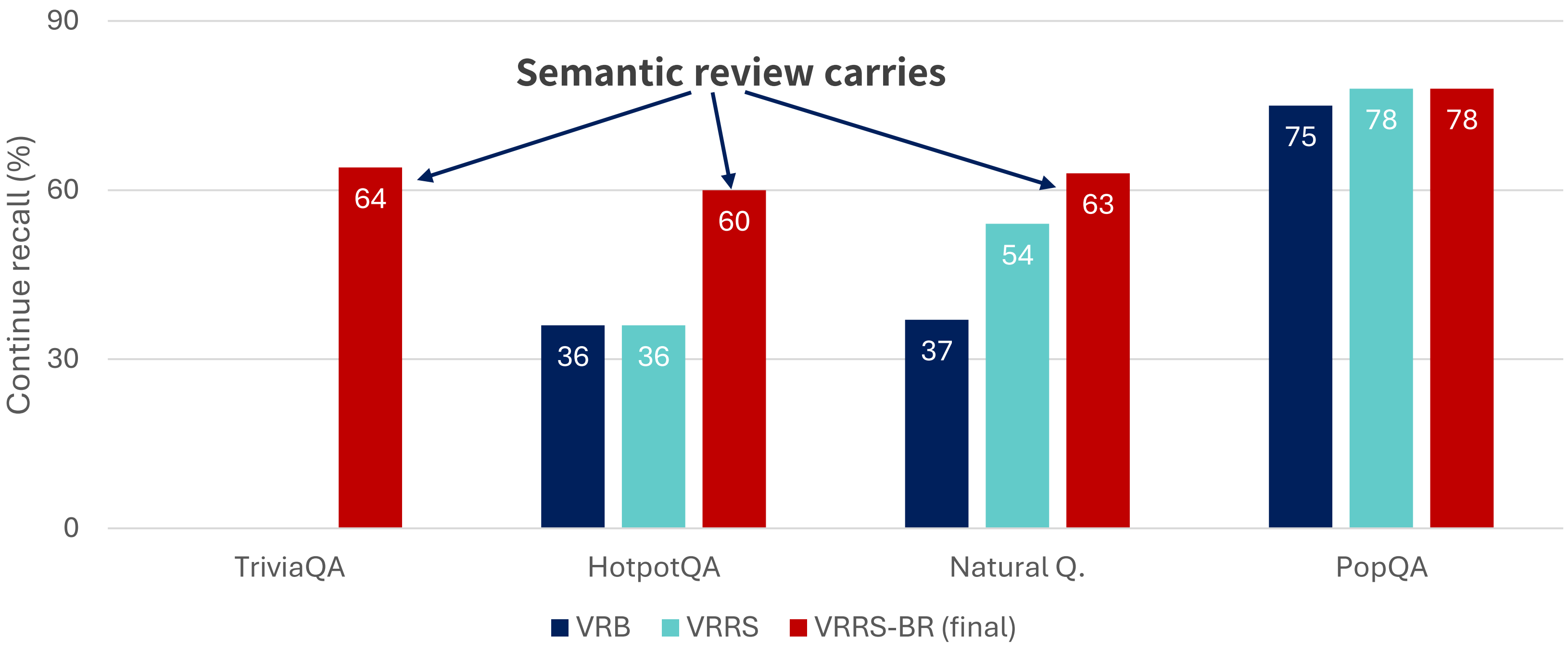
Fewer premature stops (the consequential error) for **Unnecessary continuations**.

This is a better decision, not more continuation. Continuing more often would only trade correct answers for wrong ones

VRRS-BR catches more wrong answers **while still accepting the correct ones**

The decision itself improves, not just the willingness to continue

NO SINGLE SOLUTION SURFICES: EACH POLICY OF STARC CONTRIBUTES IN DIFFERENT QA SETTINGS



Reviewer carries the gain

Fluent-but-wrong type answers in *TriviaQA* and *HotpotQA* slip past structural checks in VRB and VRRS. Semantic review catches

VRB already strong

Short-form factoid answers in PopQA are handled by VRB alone. Risk signals add little; reviewer adds nothing.

All policies add value

Complex multi-hop and ambiguous answers in Natural Questions need all policies.

The per-dataset evidence is decisive for staged design: no single stage suffices across the workload,

and **STARC is therefore irreducible rather than incidental**

STARC ARCHITECTURE GENERALISES TO OTHER AGENTIC SETTINGS

WHAT GENERALISES?	LIMITATIONS
Observable round state schema transfers to any agentic system with discrete reasoning rounds and inspectable state. **The STARC pattern** applies at any boundary where the evaluation criterion differs from the answer-generation criterion. **Aligns with adjacent work** adaptive RAG, agentic verification, process-reward modelling are converging on the same shape.	**Scale** 408 states, 26 trajectories. Bounded by A100 compute at 128K tokens (cost and time), not by design. **Judge dependence** every stage uses an LLM judge; mitigated by dual-pass labels and a manual case-study anchor. **One boundary, replay only** the first boundary, four datasets, one model. The pattern is argued to transfer, not yet tested.

RESPONSIBLE AGENTIC DEPLOYMENT

- STARC is an instance of the broader principle that agentic systems should know when to defer rather than commit.
- The same pattern applies to tool use, retrieval, and abstention decisions and should be a key consideration for all such systems

STOPPING IN AGENTIC QA IS READ FROM THE SYSTEM STATE NOT FROM MODEL CONFIDENCE

THE CONTROL POINT

Establishes **the first-round boundary as the control point**

CLARIFICATION NOT ALWAYS BENIFICIAL

Shows that continuation is not **uniformly beneficial**

Motivates the Stopping Decision

STARC ARCHITECTURE AND POLICIES

STARC, as three layered policies, to answer the Stopping Decision

Polices make better decisions, not just continue

STARC EXTENDS TO OTHER AGENTIC SYSTEMS

The idea: LLM judges of the state, composed by deterministic logic applies when an agent must decide whether to commit or continue.

Thank you for your attention

Questions welcome